DO NOW, AO1 REVISION

Nuclear fission and fusion

Atomic structure
Paper 1, Separate Physics only
12 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 7 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Define nuclear fission.**	The splitting of a large, unstable nucleus into two smaller nuclei.	☐
2	**Give two examples of elements that can undergo nuclear fission.**	Uranium and plutonium	☐
3	**What is spontaneous fission?**	Fission that happens without the nucleus absorbing a neutron.	☐
4 ★	**State what usually has to happen before a nucleus undergoes fission.**	The unstable nucleus must absorb a neutron.	☐
5 ★	**State what a nucleus splits into during fission.**	Two smaller nuclei, two or three neutrons and gamma rays	☐
6 ★	**What is a chain reaction?**	Neutrons released by one fission cause further fissions, which release more neutrons, and so on.	☐
7 ★	**Define nuclear fusion.**	The joining of two light nuclei to form a heavier nucleus.	☐
8	**In nuclear fusion, which has more mass: the two light nuclei or the nucleus they form?**	The two light nuclei	☐
9 ★	**Some mass is lost during nuclear fusion. State what it is converted into.**	Energy (of radiation)	☐
10	**Name the energy store that all fission products have.**	The kinetic energy store	☐
11 ★	**State how the chain reaction in a reactor differs from that in a nuclear weapon.**	Reactor: controlled. Weapon: uncontrolled, causing an explosion.	☐
12	**Name the process that releases energy in stars.**	Nuclear fusion	☐

panphy.app, Do Now AO1 revision, © 2026 PanPhy Labs. All rights reserved.